



Training Program 1: Advanced NLP & Large Language Models (LLMs)

- M01: Text Preprocessing
- M02: Word and Sentence Embeddings
- M03: Transformer Architectures
- M04: Entity-Centric NLP
- M05: Applied NLP Tasks
- M06: Multimodal NLP
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- M08: Prompt Engineering
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Advanced NLP & Large Language Models (LLMs)

7.1 Introduction to Large Language Models (LLMs)

7.2 LLM Scaling Laws: The Economics of AI

7.3 Foundation Model Pre-training Objectives

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7.5 Understanding and Extending Context Length

7.6 Summary and Key Takeaways

M07 LLM Fundamentals

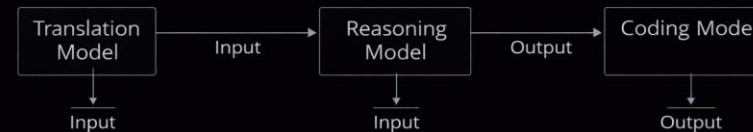
Chapter 7.1: Introduction to Large Language Models (LLMs)

What Is a Large Language Model?

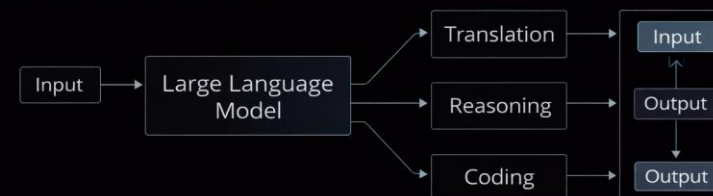
Large Language Models (LLMs) represent a shift from task-specific NLP systems to general-purpose cognitive engines. Their intelligence emerges from scale, not from handcrafted rules.

- LLMs are general-purpose models, not single-task systems
- Capabilities emerge from scale + data, not explicit programming
- Intelligence is an emergent property, not a designed feature
- One model can perform reasoning, coding, translation, and analysis

Traditional Task-Specific NLP Pipelines



Unified Large Language Model



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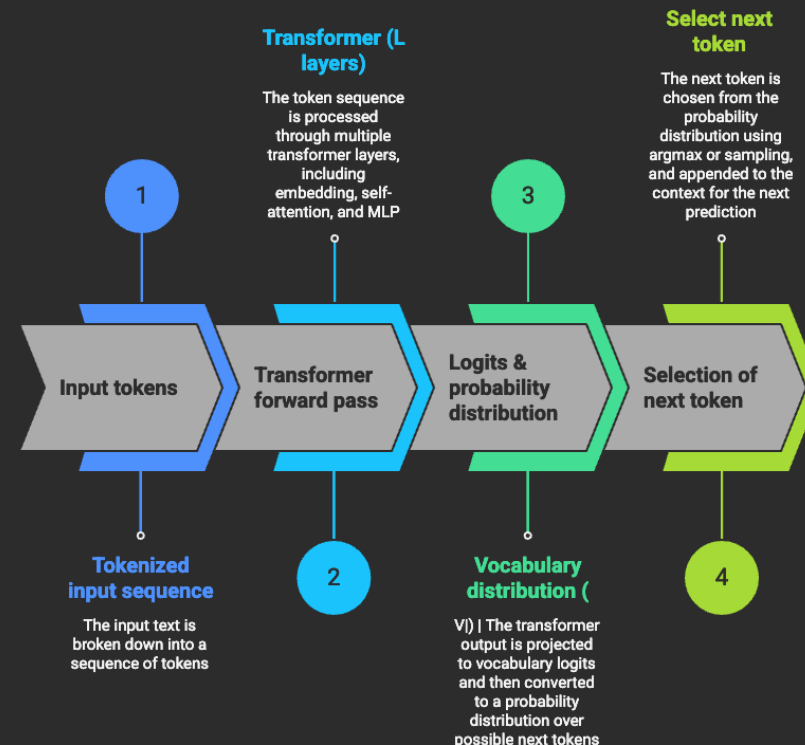
Chapter 7.1: Introduction to Large Language Models (LLMs)

The Core Mechanism: Next-Token Prediction

At its foundation, an LLM is a probabilistic system trained to predict the next token in a sequence. This simple objective enables complex reasoning when applied at massive scale.

- Core objective: predict the next token given prior context
- Model learns probability distributions over a large vocabulary
- Grammar, facts, and logic emerge from statistical learning
- No symbolic rules, only probabilistic inference

Next-Token Prediction in a Transformer Language Model



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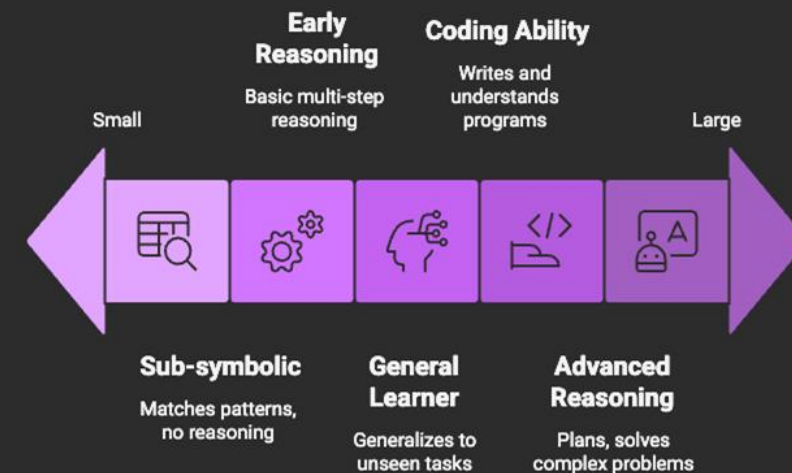
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What Makes a Model 'Large'?

LLMs are defined not just by architecture, but by quantitative scale. Parameters, data volume, and compute together determine whether advanced capabilities emerge.

- Parameter counts range from billions to trillions
- Training data includes most public human text and code
- Scale enables emergent abilities (reasoning, zero-shot learning)
- Small models cannot reliably display these behaviors

Model size influences emergent capabilities, from basic to advanced.



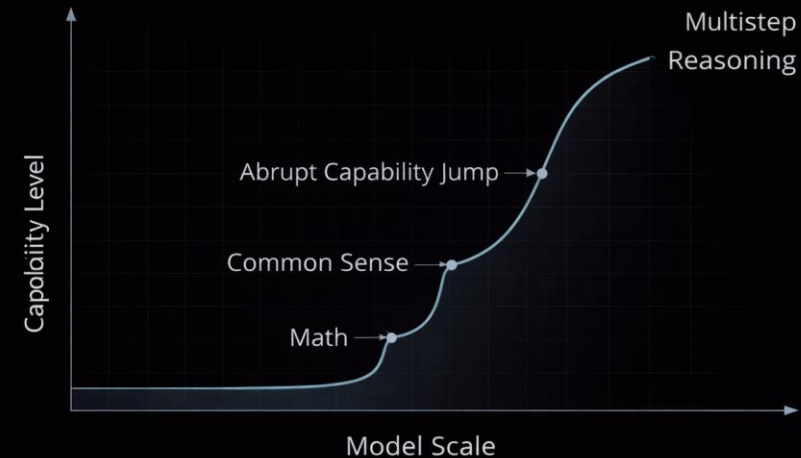
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Emergent Abilities and Threshold Effects

- Emergent abilities are non-linear effects of scale
- Examples: arithmetic reasoning, code synthesis, abstraction
- Abilities are unpredictable at small scales
- Scaling acts as a capability unlock mechanism

Emergent Abilities in Large Language Models



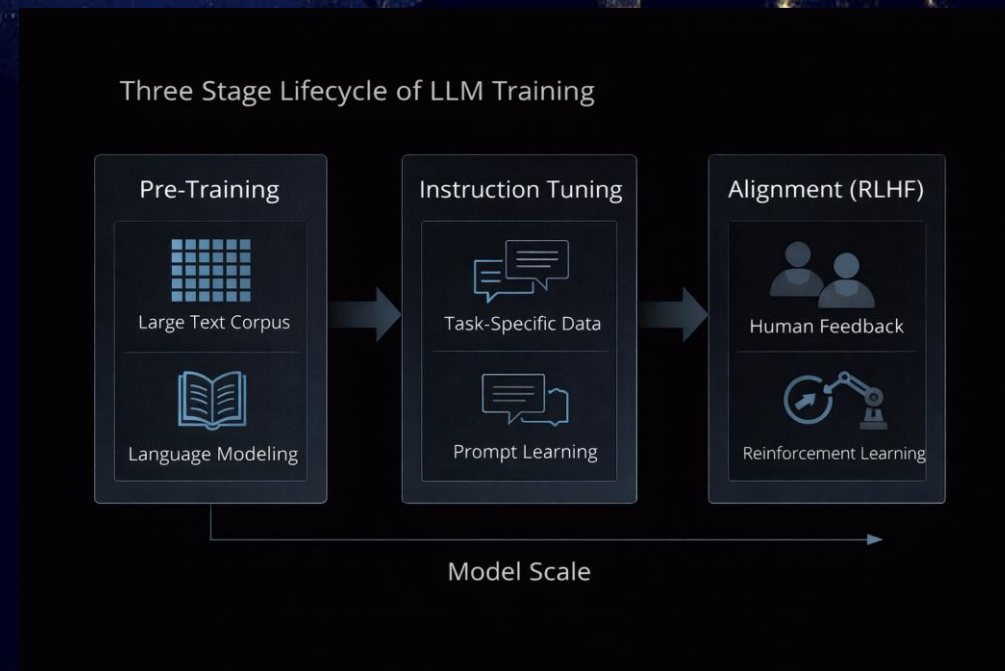
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The Lifecycle of an LLM

LLMs are not deployed directly after pre-training. They undergo a structured lifecycle that transforms raw statistical models into usable, aligned assistants.

- Pre-training: learn language and world patterns
- Instruction tuning: learn how to respond to humans
- Alignment (RLHF): optimize behavior and safety
- Each phase adds control and usefulness



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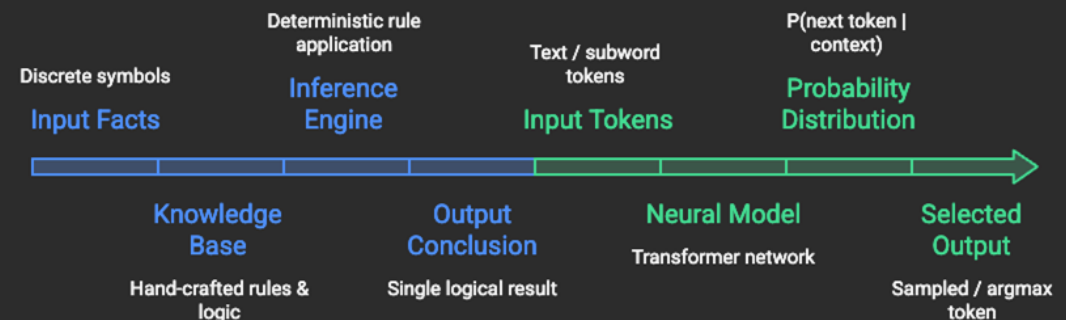
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Why This Matters Technically

Understanding what an LLM is, and what it is not, is critical for using it correctly. LLMs do not reason symbolically; they approximate reasoning through probability.

- LLMs are probabilistic, not deterministic systems
- They simulate reasoning via learned patterns
- Strengths and limits both arise from this design
- Proper usage requires understanding this foundation

Symbolic AI vs Probabilistic Language Model Reasoning





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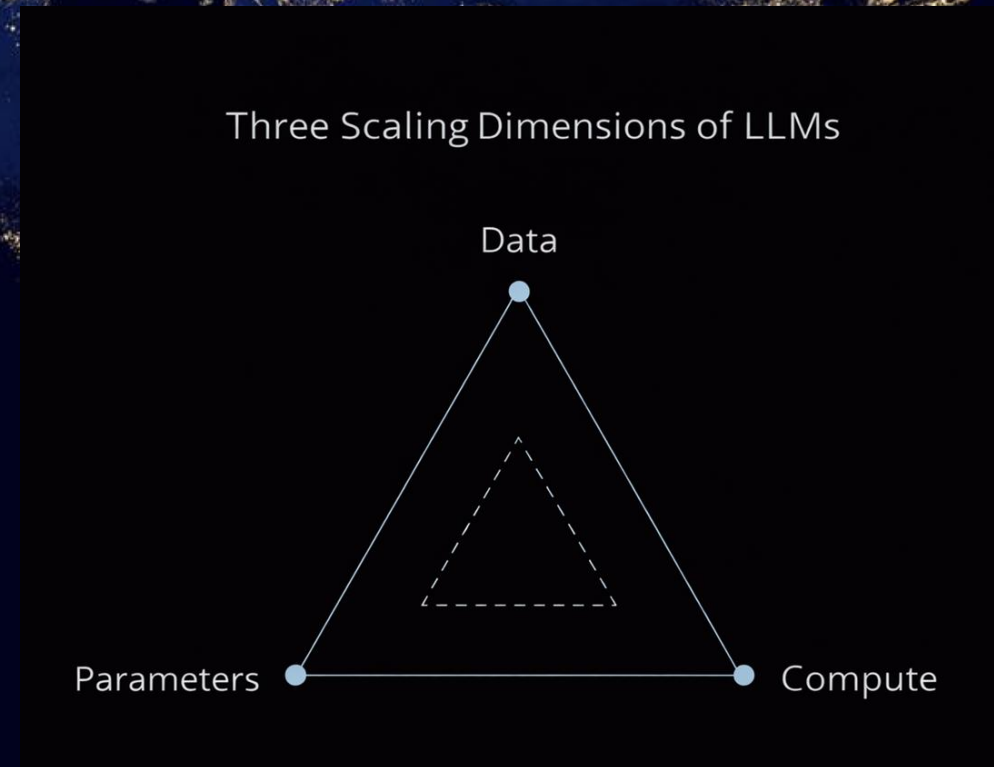
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The Three Scaling Dimensions

- LLM performance follows power-law scaling, not randomness
- Three core variables: Parameters (N), Data (D), Compute (C)
- Increasing only one factor leads to inefficient training
- Balanced scaling produces consistent loss reduction



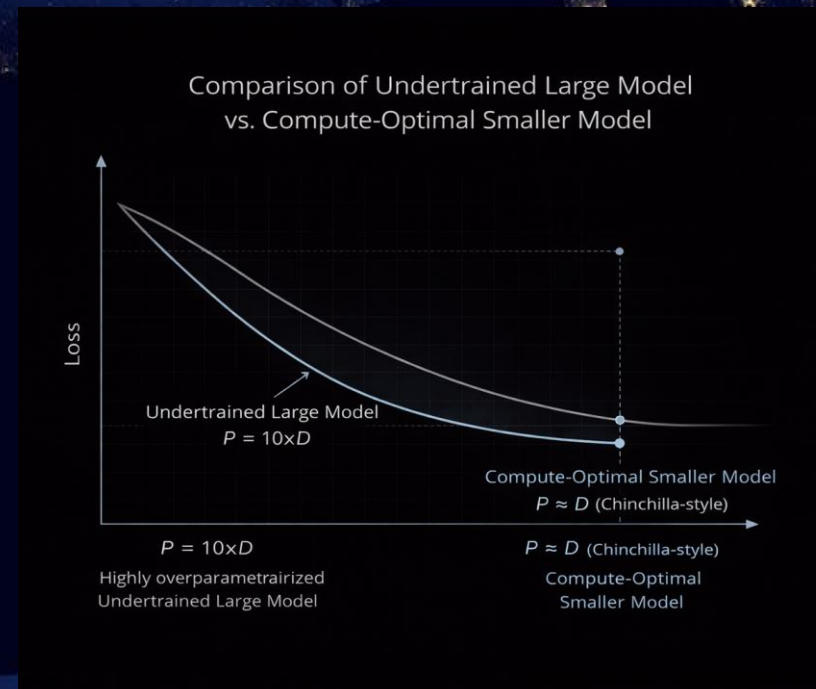
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Chinchilla and Compute-Optimal Training

Chinchilla demonstrated that many early LLMs were inefficiently trained. Optimal performance requires matching model size with sufficient training data.

- Most early models were undertrained
- Chinchilla ratio: ~20 tokens per parameter
- Smaller models + more data can outperform larger ones
- Shift toward data-efficient foundation models



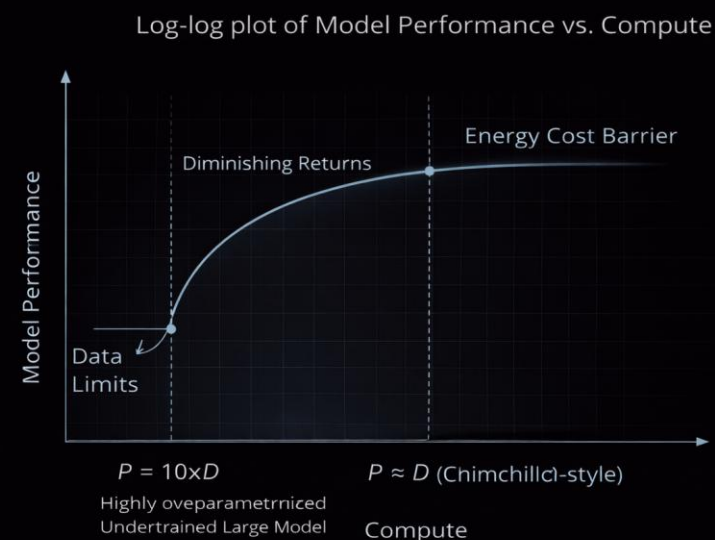
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Economic Limits and Diminishing Returns

While scaling improves intelligence, it faces physical and economic limits. Beyond certain sizes, gains become increasingly expensive and marginal.

- Scaling shows diminishing returns at trillion-parameter scale
- Constraints: data exhaustion, energy cost, hardware limits
- Compute budgets define feasible model size
- Future gains depend more on data quality and efficiency





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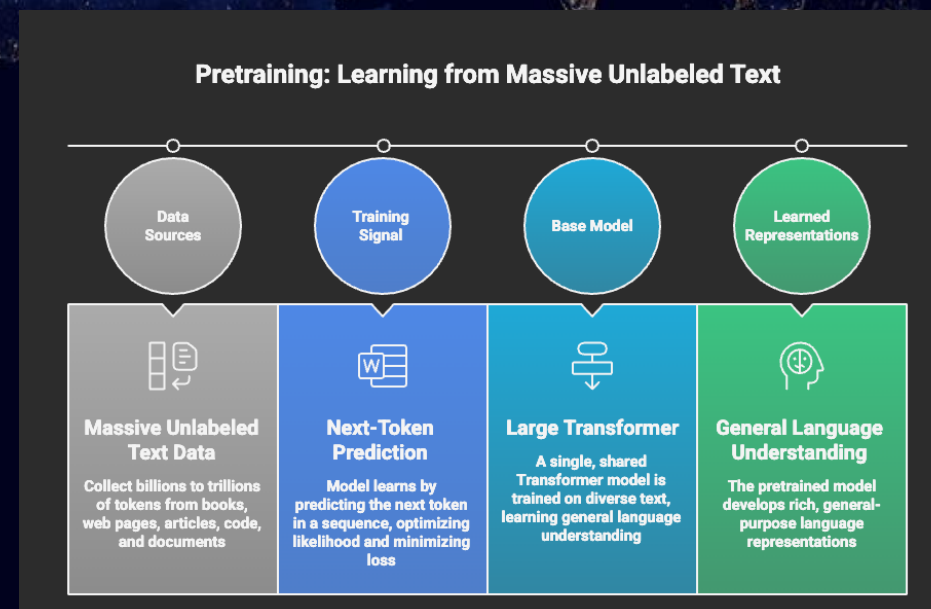
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Chapter 7.3 Foundation Model Pre-training Objectives

Purpose of Pre-training

Pre-training is the phase where an LLM acquires general intelligence from raw data. The model learns language structure and world knowledge through self-supervised learning at scale.

- Pre-training uses self-supervised learning
- No human labels: supervision comes from next-token prediction
- Objective: build a statistical model of language and facts
- Forms the base model used for all downstream tasks



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Chapter 7.3 Foundation Model Pre-training Objectives

Data Collection and Filtering

- Sources include Common Crawl, books, code, academic text
- Heavy deduplication to avoid memorization
- Removal of spam, boilerplate HTML, low-quality text
- Data diversity enables multi-domain reasoning

Data Curation Pipeline for Large-Scale Model Training



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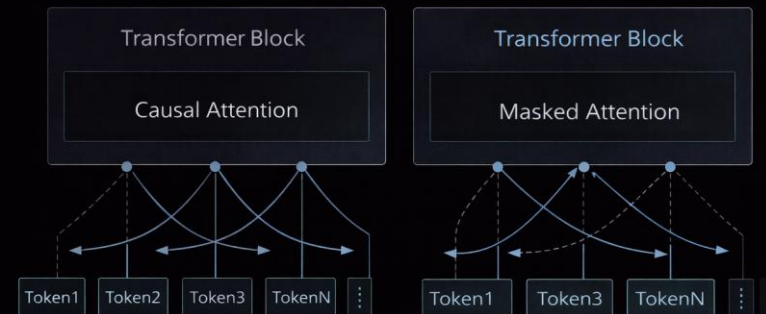
Chapter 7.3 Foundation Model Pre-training Objectives

Causal vs Masked Language Modeling

Different pre-training objectives lead to different model behaviors. Generative and understanding-focused models are trained using distinct learning strategies.

- Causal Language Modeling (CLM) → GPT-style generation
- Predicts the next token left-to-right
- Masked Language Modeling (MLM) → BERT-style understanding
- Bidirectional context but weaker for generation

Comparison of Causal Attention vs. Masked Attention



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Chapter 7.3 Foundation Model Pre-training Objectives

Tokenization as a Bottleneck

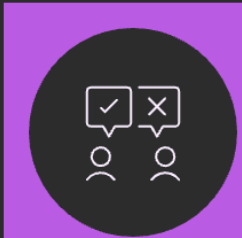
The tokenizer defines how efficiently the model perceives text. Poor tokenization wastes context and limits reasoning capacity.

- Tokenizer determines information density
- Modern models use BPE / SentencePiece
- Large vocabularies (50k–128k tokens) reduce fragmentation
- Token efficiency directly impacts context window usage

Tokenization methods

Pros

-  Efficient training
-  Faster inference
-  Handles unknown words



Cons

- Higher compute cost 
- Increased memory usage 
- More steps for model 



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Chapter 7.4 Adaptation: Fine-Tuning and Alignment

From Pre-training to Fine-Tuning

Pre-training gives models knowledge, but not behavior. Fine-tuning is the phase where a base model is transformed into a useful, task-oriented assistant.

- Pre-trained models are generic text completers
- Fine-tuning teaches how to respond, not what language is
- Uses curated instruction–response pairs
- Converts raw capability into practical utility



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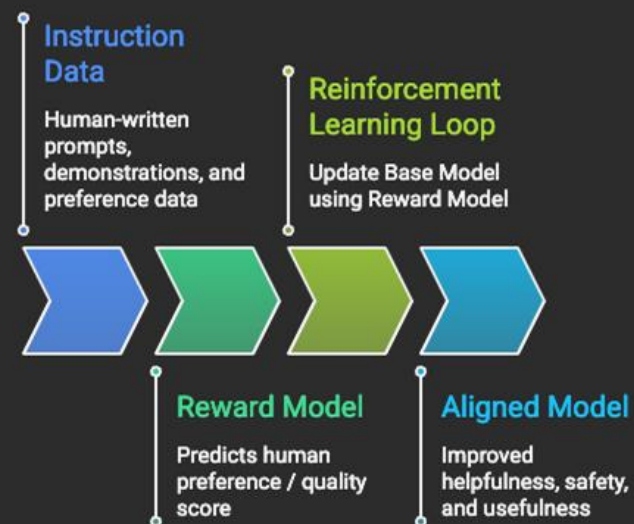
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Instruction Tuning and RLHF

Instruction tuning and human feedback align model outputs with human expectations. This step shapes tone, usefulness, and safety.

- SFT (Supervised Fine-Tuning) teaches response format
- RLHF aligns outputs with human preferences
- Humans rank answers → Reward Model is trained
- Policy optimized to be helpful, honest, and harmless

RLHF Training Pipeline



Parameter-Efficient Fine-Tuning (PEFT)

- Base model weights remain frozen
- Only a small % of parameters are trained
- LoRA is the dominant PEFT technique
- Enables fast, low-cost domain adaptation





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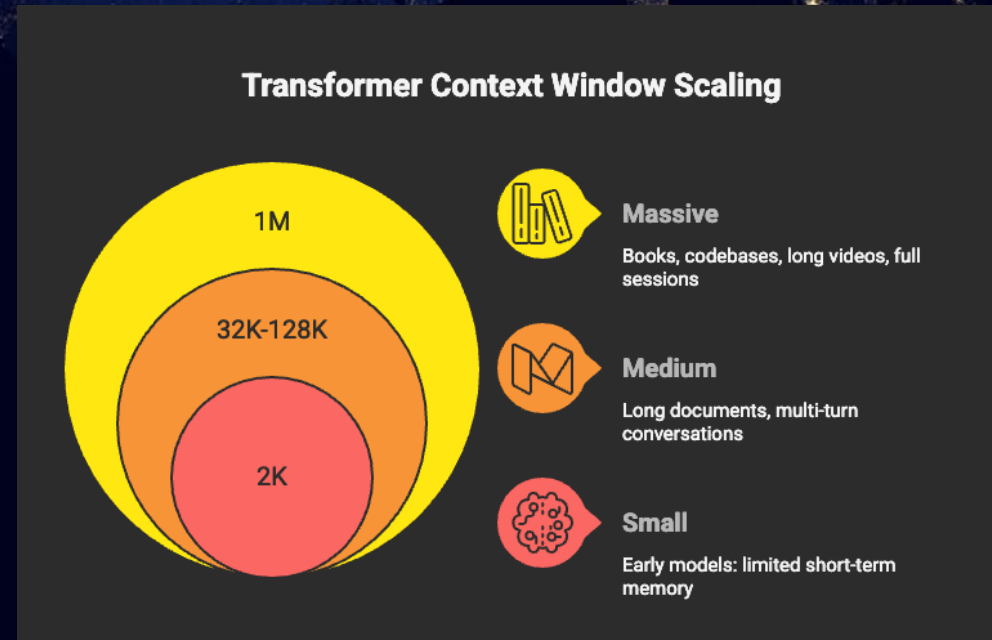
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Chapter 7.5 Understanding and Extending Context Length

Context Window and Its Limits

The context window defines how much information an LLM can “remember” at once. Extending it enables long-document reasoning but introduces severe computational constraints.

- Context window = maximum tokens per inference
- Early models: 512–2,048 tokens
- Modern models: 128k → 1M+ tokens
- Enables book-level, codebase, and video analysis



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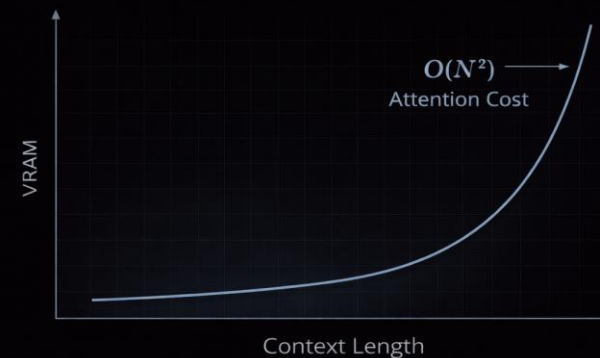
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The Quadratic Attention Bottleneck

Standard self-attention scales quadratically with sequence length. This makes long-context transformers extremely expensive in memory and compute.

- Attention complexity is $O(N^2)$
- Doubling tokens $\rightarrow 4\times$ memory usage
- Creates the “Quadratic Wall”
- Limits scalability without architectural innovation

Exponential VRAM Growth as Context Length Increases



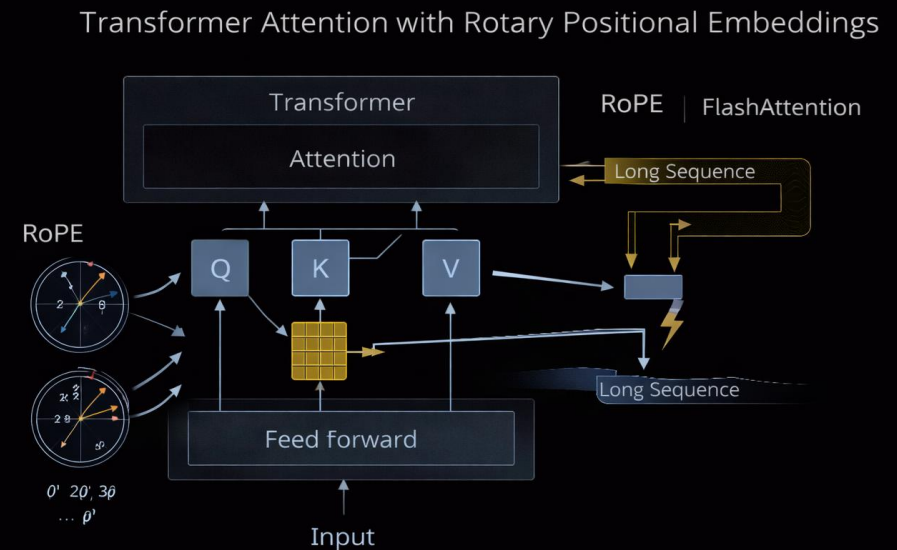
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Techniques for Long-Context Reasoning

Modern architectures bypass attention limits through algorithmic and positional innovations that preserve reasoning across long sequences.

- FlashAttention reduces memory overhead
- RoPE enables context extrapolation
- ALiBi biases attention by distance
- Solves 'Lost in the Middle' problem





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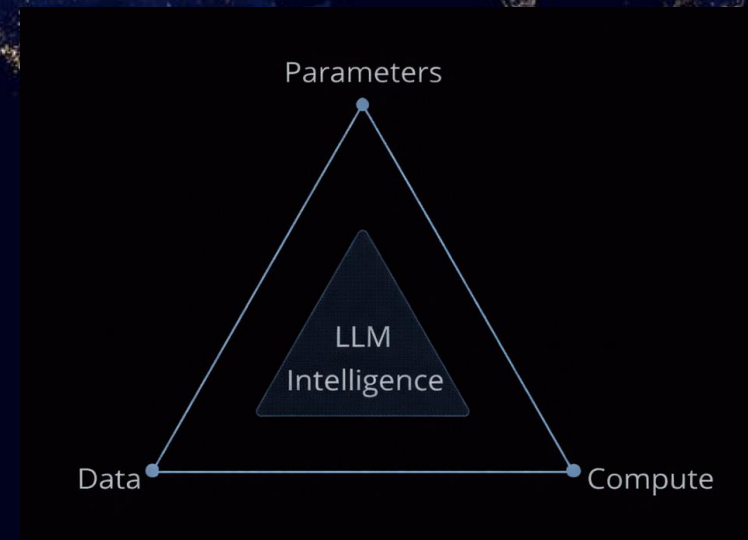
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Core Takeaways from LLM Fundamentals

Large Language Models follow predictable technical laws rather than magic. Their capabilities emerge from the structured interaction between scale, data quality, and optimization.

- Intelligence emerges from scale, not rules
- Performance depends on parameters, data, and compute
- Pre-training builds knowledge, fine-tuning builds behavior
- Context length defines reasoning depth



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Practical Trade-offs and Future Direction

Rather than being opaque or unpredictable systems, LLMs follow measurable laws driven by scale, data quality, and optimization strategies. Understanding these constraints is essential for designing reliable and economically viable AI systems.

- Trade-off between performance, cost, and latency
- Data quality sets the upper intelligence limit
- Scaling alone shows diminishing returns
- Future focus: efficiency, alignment, and control

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